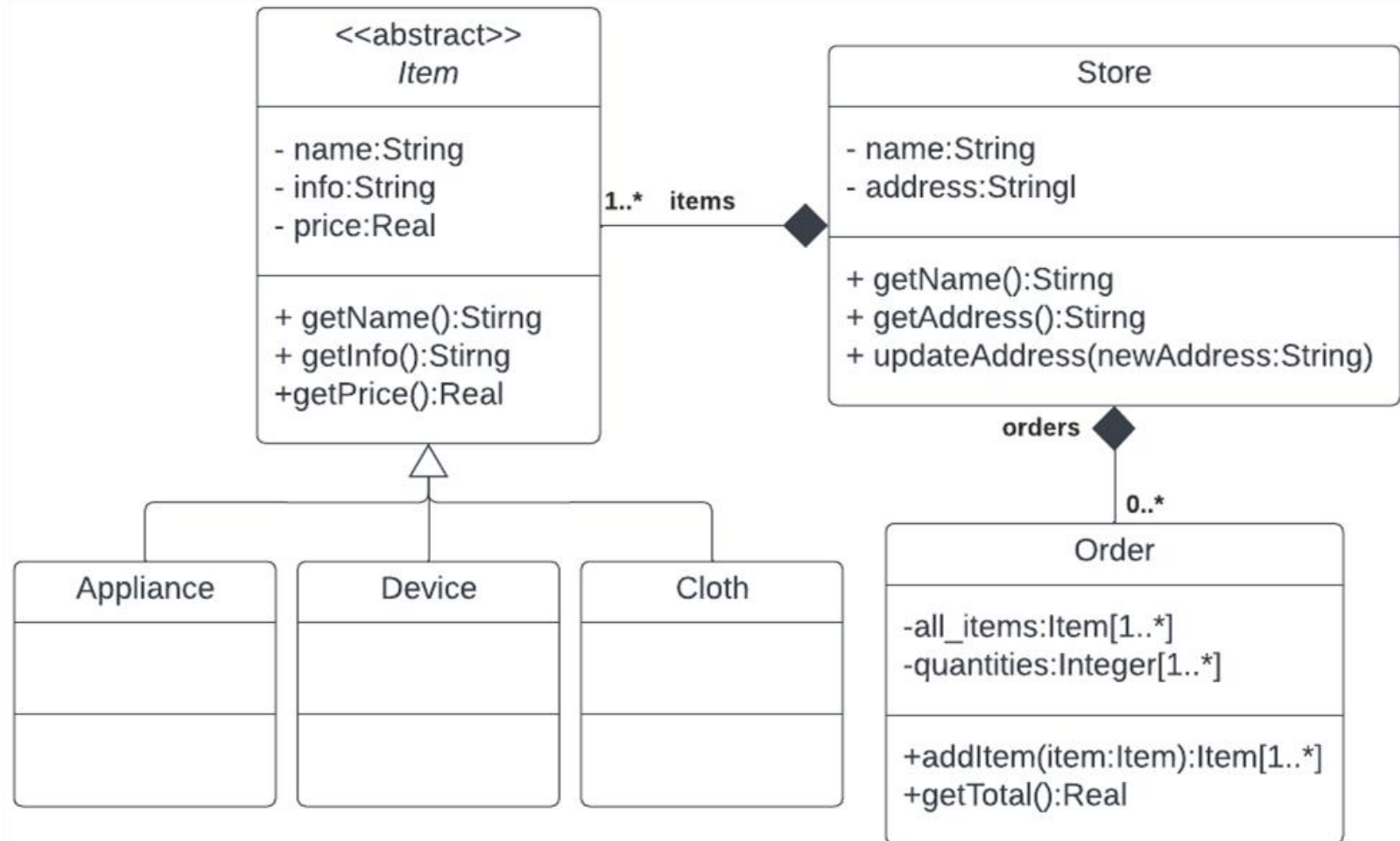


Class diagram exercise 1

- Create a class diagram that describes the structure of a store (with name and address) that contains INF but at least one item (with details name, information, and price). The store also contains INF order with basic details like list of items and their quantities.
- Items can be categorized into three types: appliance, device, and cloth
- All attributes must all be specified as private attributes
- Operations (all operations must be public)
 - Store: getName() that returns type string, updateAddress(), that accepts a new address as string
 - Item: getInfo() that returns type string, getPrice() that return real type
 - Order: addItem() that accepts a new Item and returns a List type of item, getTotal() that returns type real.

Class diagram exercise 1



Class diagram exercise 2



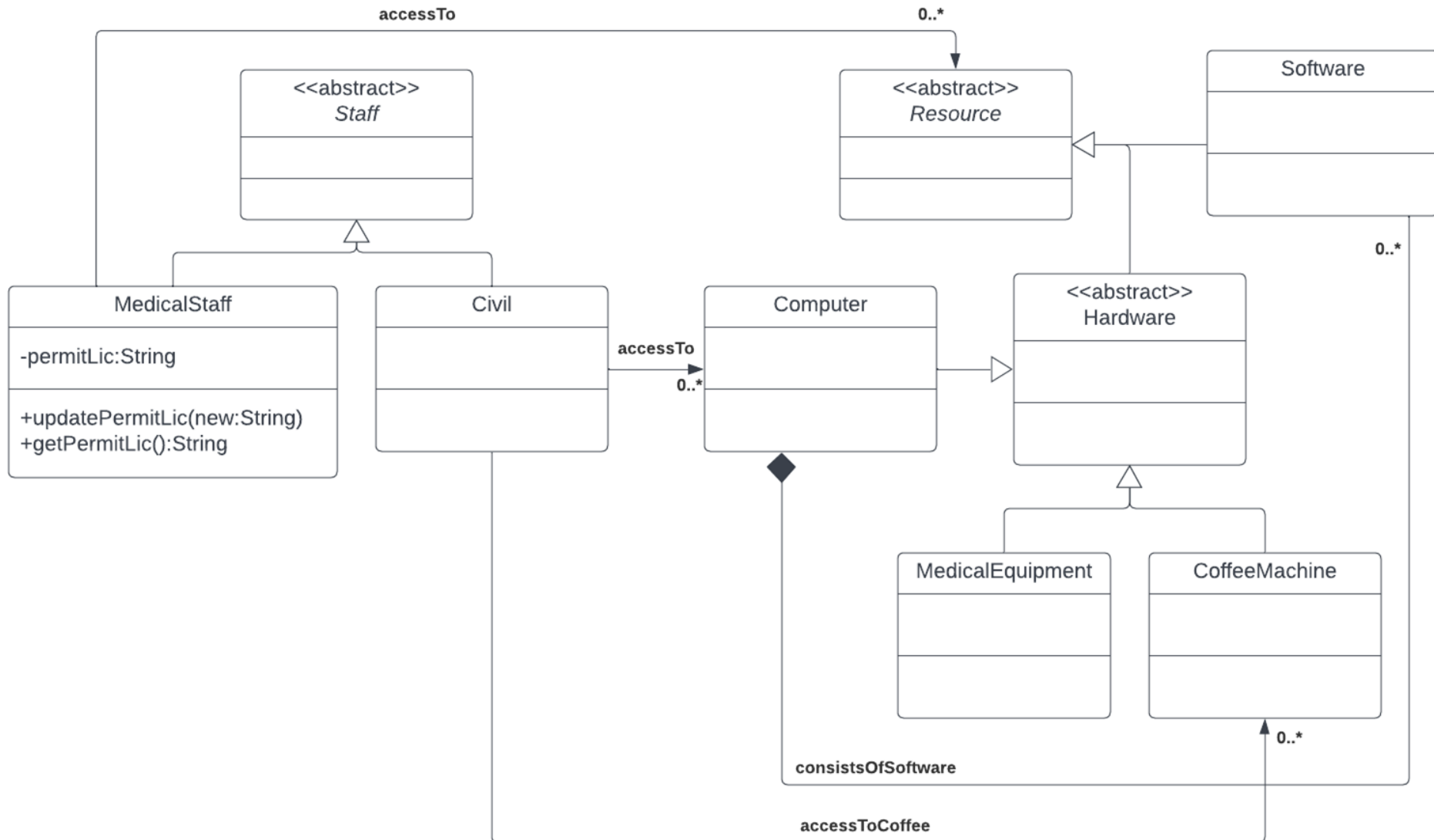
A hospital CEO wants to control hospital resources and how the staff of the hospital access them. He/she hires you and wants to construct a model structure that describes your proposed solution. There are many resources and they can be group into two types (although they can have common properties):

- Hardware: computers, medical equipments and coffee machines
- Software: computer applications

The staff can be medical staff or civil staff. Medical staff has access to everything including medical equipments, which is done using medical permits that can be renewed annually. Civil employee can access everything except medical equipments.

A computer hardware can consists of a number of INF software

Class diagram exercise 2



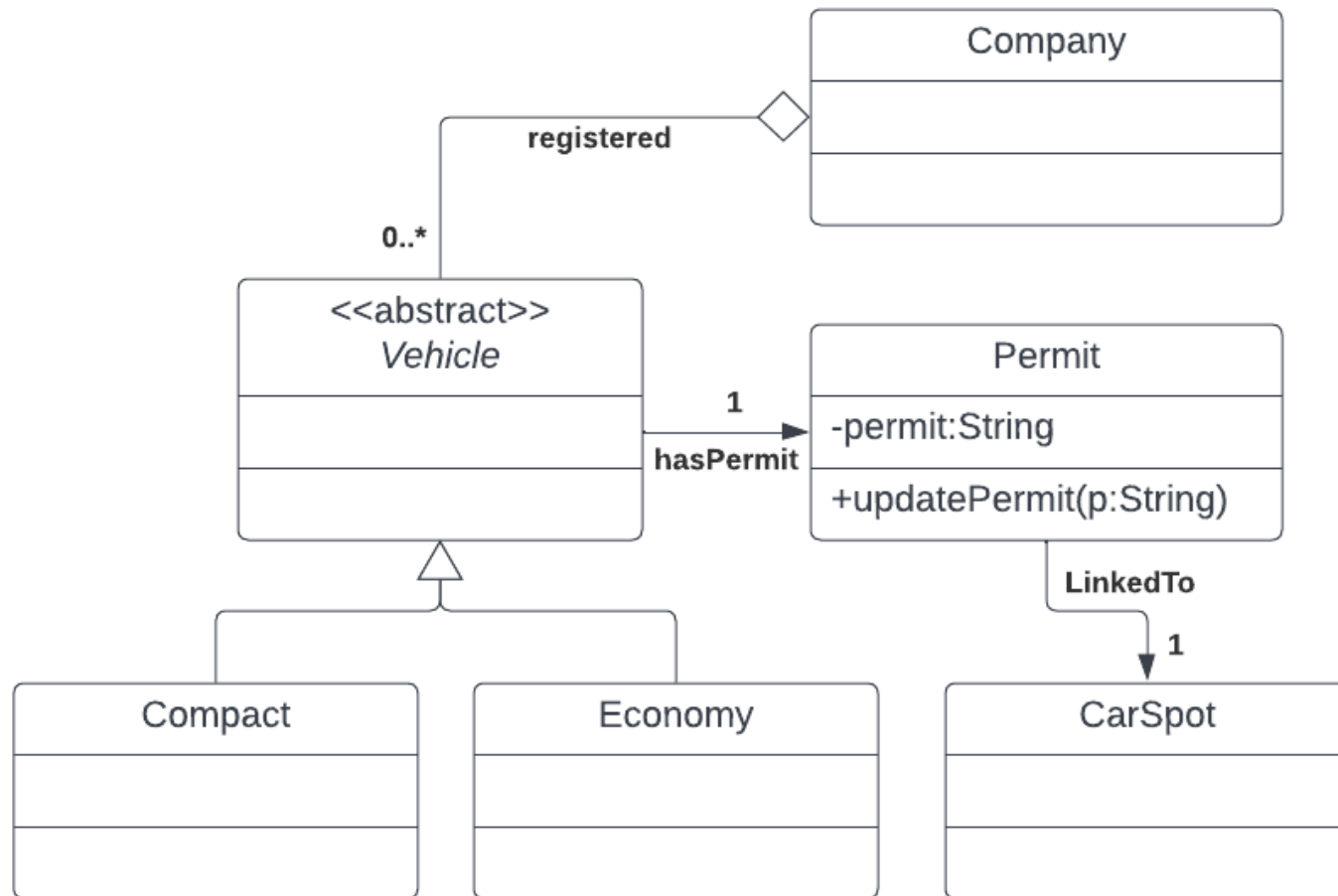
Class diagram exercise 3



A company needs to register all its employees' personal transportations that they are using to come work. The company needs to track parking permits provided to vehicles. There are two types of vehicles allowed, compact and economy cars. Each registered car has a permit linked to it and there is always one parking spot for each permit. Permits are encoded into a string and each parking spot has a number.

Any permit can be update by calling the function `updatePermit(p)`.

Class diagram exercise 3



Class diagram exercise 4



You are to model a representation of a tree data structure of objects. There are two types of trees: binary and nonBinary, but they share a common attribute 'label' of the object type, with 'null' as the default value.

A binary tree may have two objects of binary trees: firstChild and secondChild. The multiplicity of these two associations must be optional.

A nonBinary tree has zero or more objects of nonBinary trees, and this is represented by an association named 'children'.

An object class can be either a StringObject class with a string attribute named 'value' or a RealObject class with a real attribute named 'value'.

Provide a class diagram that illustrates the above structure. Identify the appropriate relationships between the classes. All attributes of classes must be made public.

Class diagram exercise 4

